

HDOC Day-27

Name	Gurveer Singh
SAF ID	590032925
Batch	15

Problem Statement

Write a menu-driven C program using a switch statement to print the selected Pyramid/Diamond pattern. Use nested loops without arrays. Each pattern is implemented in a separate parameterless function with no return value. Prepare the algorithm and test-case table, then compile, execute and test the program.

Menu Options

1. Full pyramid
2. Inverted pyramid
3. Diamond pattern
4. Palindromic number pyramid
5. Repeated-number pyramid
6. Exit

Algorithm

1. Start the program.
2. Display the menu inside a loop.
3. Read the user's choice.
4. Use switch-case to call the selected pattern function.
5. Inside the function, read the number of rows n and use nested loops to print the pattern.
6. For an invalid choice, display an error message.
7. Repeat the menu until the user selects Exit.
8. Stop.

C Program

```
#include <stdio.h>

void fullPyramid() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("*");
        printf("\n");
    }
}

void invertedPyramid() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = n; i >= 1; i--) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("*");
        printf("\n");
    }
}

void diamond() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
```

```

        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("*");
        printf("\n");
    }
    for (i = n-1; i >= 1; i--) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("*");
        printf("\n");
    }
}

void palindromePyramid() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= i; j++) printf("%d ", j);
        for (j = i-1; j >= 1; j--) printf("%d ", j);
        printf("\n");
    }
}

void repeatedPyramid() {
    int n, i, j;
    printf("Enter number of rows: ");
    scanf("%d", &n);
    for (i = 1; i <= n; i++) {
        for (j = 1; j <= n-i; j++) printf(" ");
        for (j = 1; j <= 2*i-1; j++) printf("%d ", i);
        printf("\n");
    }
}

int main() {
    int choice;
    do {
        printf("\n1. Full pyramid\n");
        printf("2. Inverted pyramid\n");
        printf("3. Diamond pattern\n");
        printf("4. Palindromic number pyramid\n");
        printf("5. Repeated-number pyramid\n");
        printf("6. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: fullPyramid(); break;
            case 2: invertedPyramid(); break;
            case 3: diamond(); break;
            case 4: palindromePyramid(); break;
            case 5: repeatedPyramid(); break;
            case 6: printf("Program exited.\n"); break;
            default: printf("Invalid choice. Try again.\n");
        }
    } while (choice != 6);
    return 0;
}

```

Test-Case Table

Test	Choice	n	Expected result
1	1	4	4-row full pyramid
2	2	4	4-row inverted pyramid
3	3	4	7-row diamond
4	4	4	Palindromic number pyramid
5	5	4	Repeated-number pyramid

Test	Choice	n	Expected result
6	7	-	Invalid choice message
7	6	-	Program exits

Sample Output (n = 4)

Choice 1 - Full Pyramid

```
Enter your choice: 1
Enter number of rows: 4
*
**
***
****
*****
```

Choice 2 - Inverted Pyramid

```
Enter your choice: 2
Enter number of rows: 4
*****
****
***
**
*
```

Choice 3 - Diamond Pattern

```
Enter your choice: 3
Enter number of rows: 4
*
**
***
****
*****
*****
****
***
**
*
```

Choice 4 - Palindromic Number Pyramid

```
Enter your choice: 4
Enter number of rows: 4
1
1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
```

Choice 5 - Repeated-Number Pyramid

```
Enter your choice: 5
Enter number of rows: 4
1
2 2 2
3 3 3 3 3
4 4 4 4 4 4 4
```